2 **(a)** energy (stored) / work done represented by area under graph **B1**

energy = ½ (180) (4.0×10^{-2}) **C1**

= 3.6 J **A1**

(b) (i) *either* momentum before release is zero **M1**

so sum of momenta of trolleys after release is zero **A1**

or force = rate of change of momentum **M1**

force on trolleys equal and opposite **A1**

or impulse = change in momentum **M1**

impulse on each trolley is equal and opposite **A1**

(ii) 1. $M_1V_1 = M_2V_2$ **B1**

2. $E = \frac{1}{2} M_1V_1^2 + \frac{1}{2} M_2V_2^2$ **B1**

(iii) 1.

$$E_K = \frac{1}{2}mv^2$$

$$= \frac{1}{2}\left(\frac{m}{m}\right)mv^2$$ **M1**

$$= \frac{(mv)^2}{2m}$$

$$= \frac{p^2}{2m}$$ **A0**

2. As trolley B has a smaller mass, it has a larger kinetic energy <u>because momentum is the same/ constant for both trolleys</u>. **B1**

3 **(a)** Phase difference = $\frac{2\pi t}{\;} = \frac{2\pi(2.5–1.5)}{\;} = \frac{2\pi}{\;} = 120^{\circ}$ **A1**